FROM TRANSMISSION-LINE MODELS TO STUART–LANDAU OSCILLATORS: A HOPF-BASED DESCRIPTION OF COCHLEAR DYNAMICS

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Overview

ANALYTICAL REDUCTION: Transmission-line element → Hopf normal form

OSCILLATOR REPRESENTATION: Cochlear partition → coupled Stuart -Landau Oscillators (SLOs)

LARGE-SCALE SIMULATION: 600 oscillators · 200 Hz–16 kHz

COCHLEAR CHARACTERISTICS: Frequency selectivity, level-dependent compression and half-octave shift

EMERGENT DYNAMICS: Traveling-wave-like propagation

Analytical Reduction

Local Transmission Line Model

$$v_t = \dot{x}_t$$

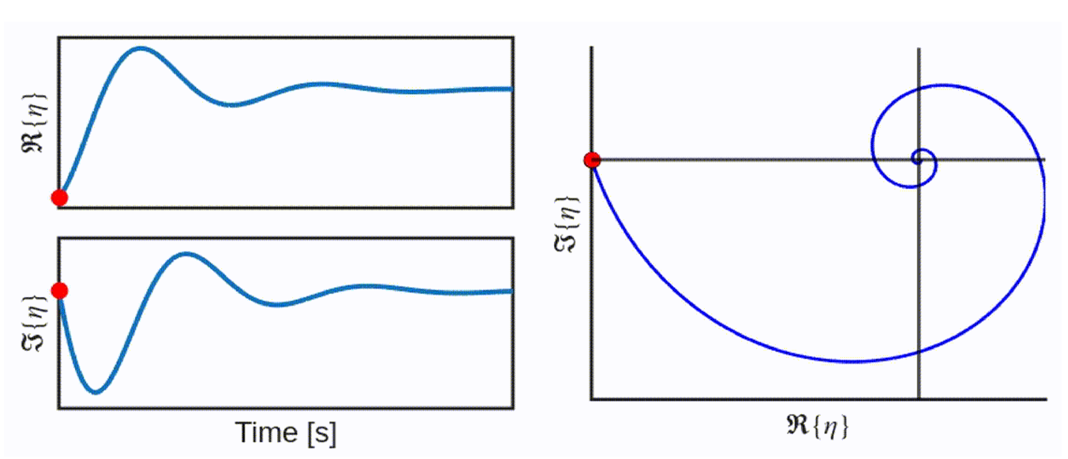
$$\dot{v}_t = -\frac{k}{m}x_t - \frac{d}{m}v_t - \frac{1}{m}(\Psi(0) + \dot{\Psi}(0)x_{t-\tau} + \frac{\ddot{\Psi}(0)}{2}x_{t-\tau}^2)x_{t-\tau}$$

Stuart Landau Oscillator

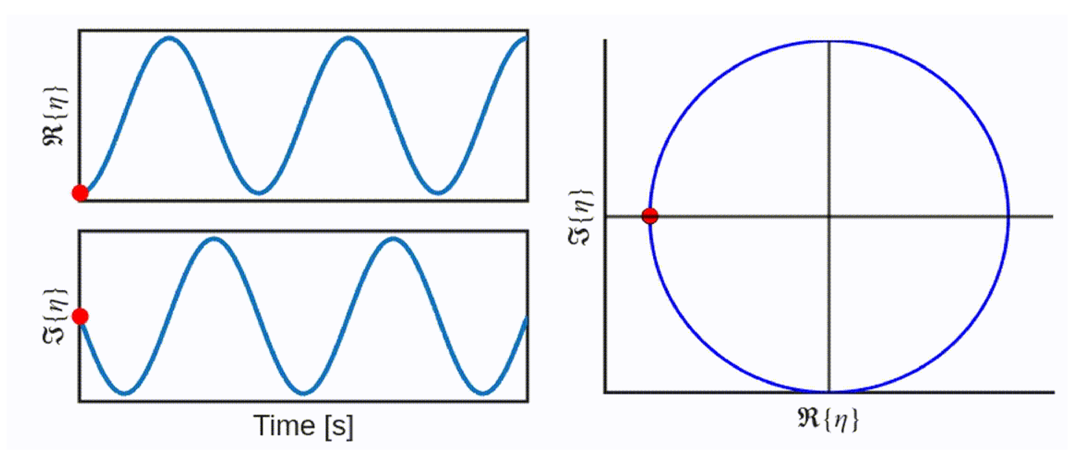
$$\eta_t = x_t - j\frac{1}{\omega}v_t$$

$$\dot{\eta}_t = (\mu + j\omega_0)\eta_t - (\alpha + j\beta)\left|\eta_t\right|^2\eta_t$$

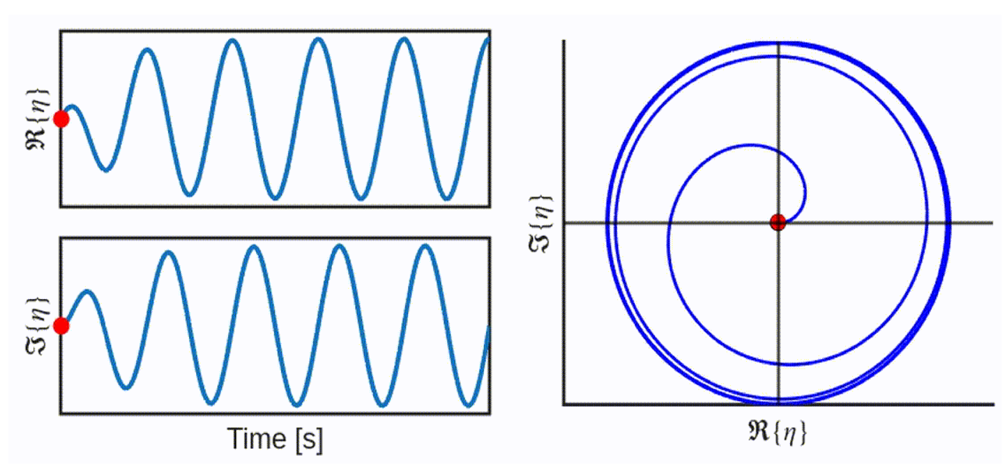
$$\mu < 0; \quad \omega\tau = \frac{\pi}{2}$$



$$\mu = 0; \quad \omega\tau = 0$$



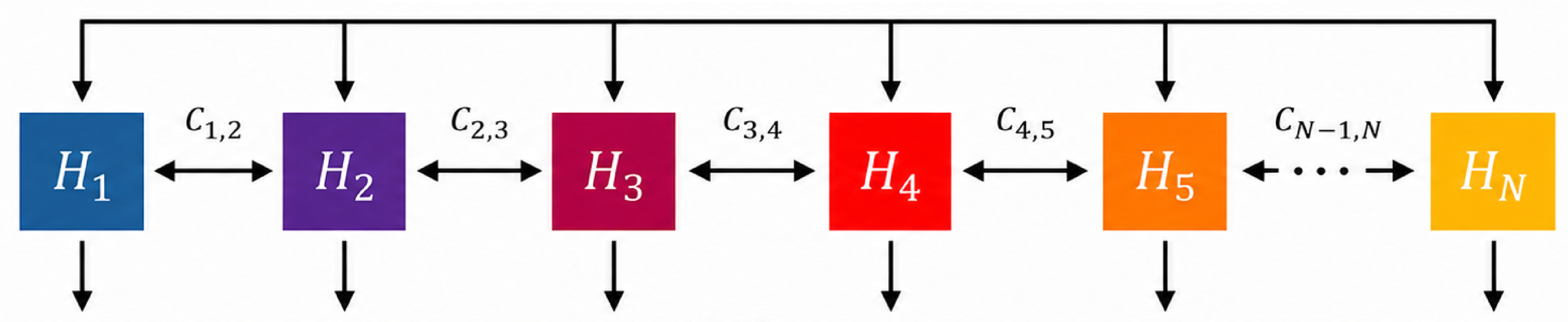
$$\mu > 0; \quad \omega\tau = \frac{3\pi}{2}$$



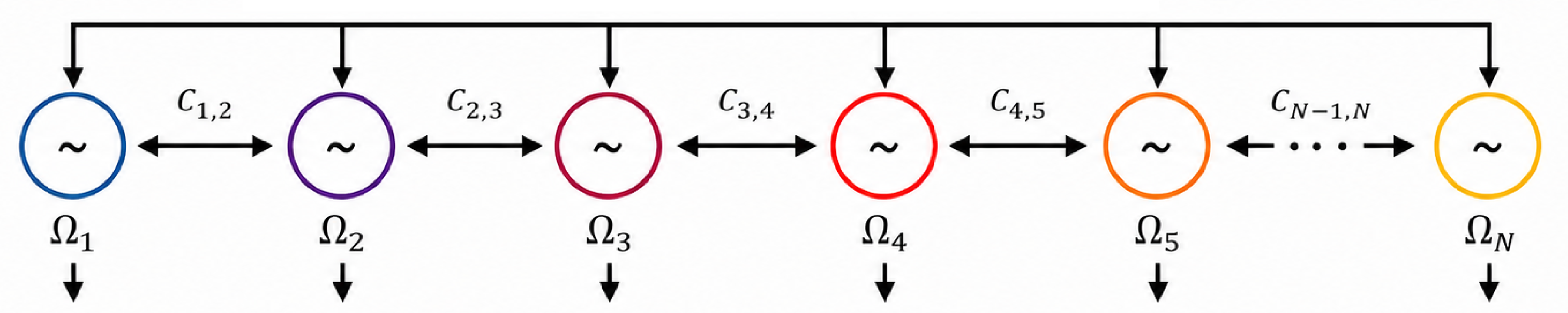
x_t	:	Displacement	v_t	:	Velocity	η_t	:	Complex oscillator state	Ψ	:	Active-feedback function
m	:	Effective mass	k	:	Stiffness	d	:	Damping coefficient	ω_0	:	Natural angular frequency
μ	:	Linear dynamics parameter	α	:	Nonlinear dynamics parameter	β	:	Asynchronicity parameter			

Oscillator Representation

Transmission-Line Model

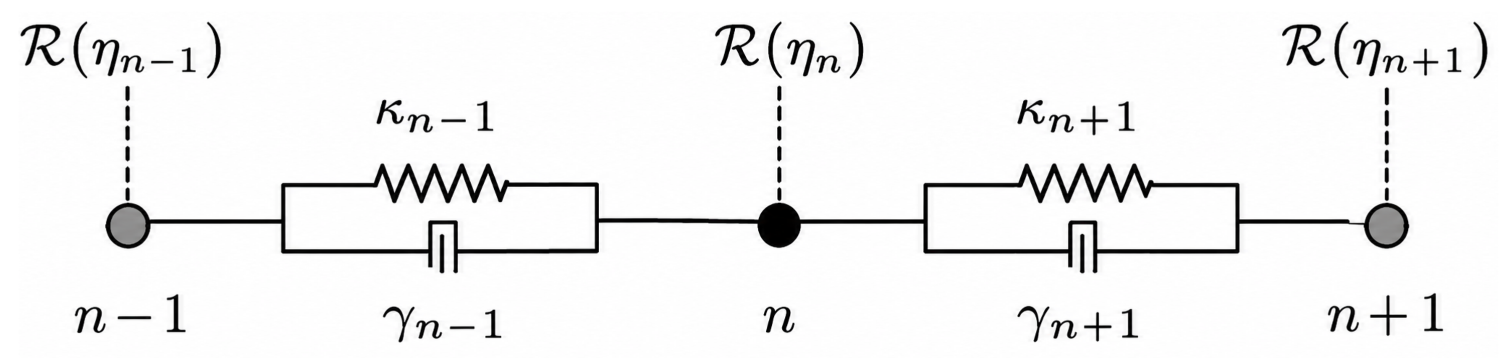


Oscillator Bank



Membrane (Nearest Neighbor) Coupling

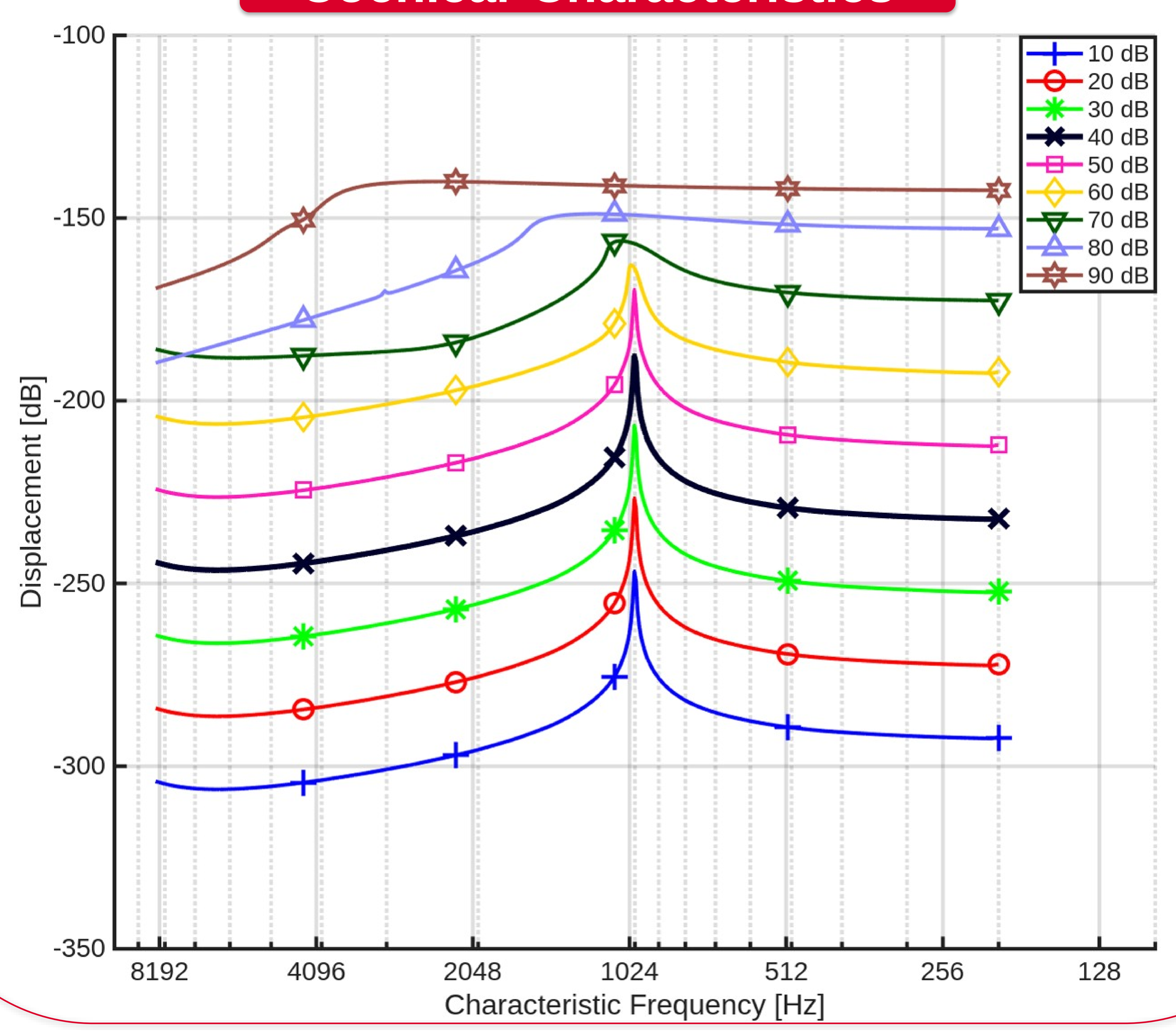
$$C(\eta) = \kappa_{n-1}\mathcal{R}(\eta_{n-1}) + \kappa_{n+1}\mathcal{R}(\eta_{n+1}) - j\left[\gamma_{n-1}\mathcal{I}(\eta_{n-1}) + \gamma_{n+1}\mathcal{I}(\eta_{n+1})\right].$$



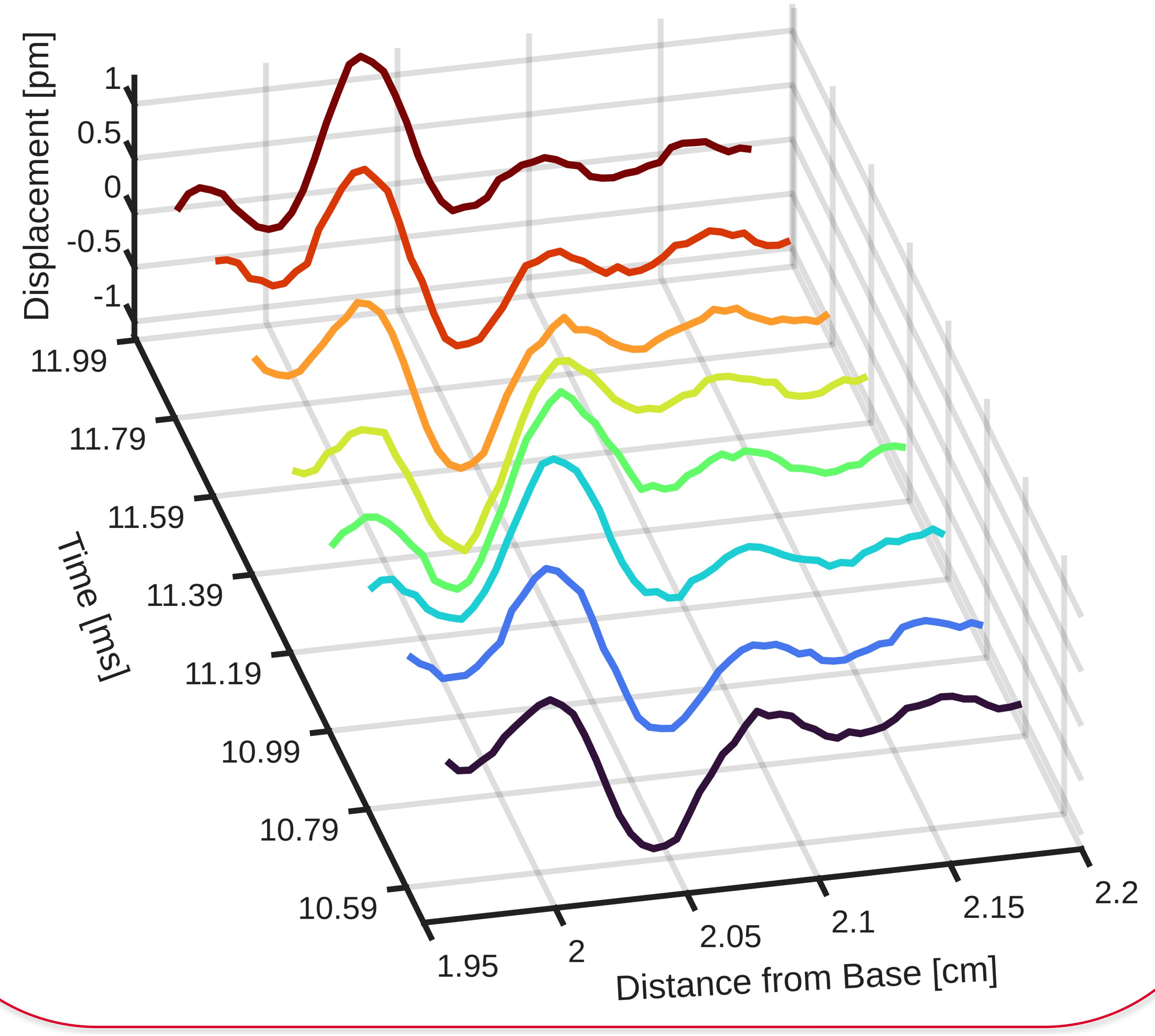
$$\kappa_{n\pm1} \in \mathbb{R} : \quad \text{nearest-neighbor stiffness coupling,}$$

$$\gamma_{n\pm1} \in \mathbb{R} : \quad \text{nearest-neighbor damping coupling.}$$

Cochlear Characteristics



Emergent Dynamics



Conclusion

Local transmission-line dynamics can be reduced to a Stuart–Landau oscillator near the Hopf bifurcation. An array of coupled SLOs preserves key cochlear characteristics—frequency selectivity, nonlinear compression, half-octave shift, and traveling-wave-like dynamics—providing a simpler dynamical representation of the cochlear partition.